

Jieyin Yang

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EDUCATION

University of Chinese Academy of Sciences

Aug. 2021 – present

Ph.D. in Applied Mathematics

Advisor: Prof. [Xiaohong Jia](#)

GPA: 3.91/4.0

Columbia University

Jan. 2020 – Jun. 2020

Visiting Student in Engineering and Applied Sciences

University of Chinese Academy of Sciences

Aug. 2017 – Jun. 2021

B.S. in Mathematics and Applied Mathematics

GPA: 3.95/4.0, Rank: 2/52

RESEARCH INTERESTS

Computer Graphics, Computer Aided Design, Computational Algebraic Geometry

PUBLICATIONS

8. A Robust and Efficient Intersection Algorithm for NURBS Surfaces: Handling Small Loops and Tangent Intersections.

Jieyin Yang, Xiaohong Jia

ACM Transactions on Graphics, 2026

7. Advances and Challenges in Surface–Surface Intersection Computation — An Overview.

Kai Li, **Jieyin Yang**, Xiaohong Jia

Computer-Aided Design, 2026

6. Overlap Region Extraction of Two NURBS Surfaces.

Jieyin Yang, Xiaohong Jia

ACM Transactions on Graphics (Proc. SIGGRAPH Asia 2025)

5. Improving the Watertightness of Parametric Surface/Surface Intersection.

Yuqing Wang, Xiaohong Jia, **Jieyin Yang**, Bolun Wang, Pengbo Bo, Yang Liu

Computer Graphics Forum, 2025

4. Boolean Operation for CAD Models Using a Hybrid Representation.

Yingyu Yang, Xiaohong Jia, Bolun Wang, **Jieyin Yang**, Shiqing Xin, Dong-Ming Yan

ACM Transactions on Graphics (Proc. SIGGRAPH 2025)

3. Computing the Intersection of Two Ellipsoids Based on a Fast Algebraic Topology Determination Strategy.

Xiao Chu, Kai Li, Xiaohong Jia, **Jieyin Yang**, Jiarui Kang

Computer Aided Geometric Design (Proc. GMP 2025)

2. Accurate and Robust Registration of Low Overlapping Point Clouds.

Jieyin Yang, Mingyang Zhao, Yingrui Wu, Xiaohong Jia
Computer & Graphics, 2024

1. Topology Guaranteed B-Spline Surface-Surface Intersection.

Jieyin Yang, Xiaohong Jia, Dong-Ming Yan
ACM Transactions on Graphics (Proc. SIGGRAPH Asia 2023)

HONORS AND AWARDS

2025	CSIAM Applied Mathematics Achievement Award: Surface Intersection Method and Software for Geometry Engine
2024	Doctoral National Scholarship
2023	Hua Luogeng Scholarship of AMSS
2021-25	Merit Student of CAS
2020	Undergraduate National Scholarship

SKILLS

Programming	C++, Matlab, Maple
Math	Computational geometry, Symbolic computation, Linear algebra, Numerical optimization